

PORTO SEGURO, Brazil

WMO: 834600

Lat: 16.439S

Lon: 39.081W

Elev: 51

StdP: 100.71

Time Zone: -3.00 (W03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	17.8	14.8	10.6	20.0	15.8	11.3	20.2	7.4	23.9	6.7	24.4	2.5	270	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	31.1	25.0	30.2	24.7	29.9	24.6	25.9	29.1	25.6	28.8	25.3	28.4	6.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.2	27.6	24.2	19.3	26.9	24.2	19.2	26.9	80.2	29.4	78.6	28.8	77.6	28.5	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.2	7.3	15.8	32.7	1.4	0.8	14.8	33.3	14.0	33.8	13.2	34.2	12.2	34.8		
(5)			15.1	26.8	1.6	0.9	14.0	27.4	13.0	28.0	12.2	28.5	11.0	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	25.8	26.0	26.1	25.2	23.9	23.0	22.2	22.1	22.9	24.0	24.8	25.6		
	DBStd	1.91	1.01	1.05	1.16	1.31	1.31	1.40	1.15	1.29	1.43	1.43	1.54	1.18		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDD10.0	5214	489	448	498	457	431	389	378	376	387	435	443	484		
	CDD18.3	2173	231	215	239	207	172	139	119	118	137	177	193	226		
	CDH23.3	14458	1961	1987	2075	1547	926	557	381	399	623	1013	1232	1757		
	CDH26.7	2796	451	512	547	337	119	41	10	12	43	121	216	387		
(14) Wind	WSAvg	3.3	3.7	3.5	3.3	2.8	2.6	2.6	2.8	3.1	3.7	4.0	3.9	4.0		
(15) Precipitation	PrecAvg	1767	96	102	125	197	168	136	173	104	147	169	139	143		
	PrecMax	2307	290	250	236	323	371	293	487	224	370	346	438	321		
	PrecMin	1005	10	11	10	64	10	42	49	41	51	41	26	13		
	PrecStd	370	80	72	57	75	83	73	102	49	76	90	109	89		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	32.0	32.0	31.1	30.0	29.1	27.9	28.0	28.9	29.9	30.8	31.2		
		MCWB	24.9	25.4	25.6	25.4	24.5	23.9	22.9	22.7	23.3	24.1	24.7	24.7		
	2%	DB	30.1	30.9	30.9	30.1	28.9	27.8	26.2	26.9	27.9	28.9	29.8	30.1		
		MCWB	24.7	25.0	25.2	25.0	24.0	23.4	22.1	22.3	23.0	23.8	24.2	24.4		
	5%	DB	29.4	30.0	30.1	29.2	28.0	27.0	26.0	26.1	27.1	28.0	28.9	29.2		
		MCWB	24.5	24.7	24.8	24.6	23.7	22.9	21.9	21.7	22.6	23.4	24.1	24.2		
	10%	DB	29.0	29.1	29.2	28.8	27.1	26.1	25.2	25.2	26.2	27.1	28.0	28.9		
		MCWB	24.4	24.5	24.6	24.4	23.2	22.5	21.5	21.3	22.1	23.0	23.7	24.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	26.2	26.2	26.2	25.3	24.7	23.6	23.5	24.1	25.2	25.6	25.9		
		MCDB	29.2	29.5	29.6	29.4	28.4	27.5	26.6	26.7	27.7	28.4	29.1	29.0		
	2%	WB	25.5	25.7	25.9	25.7	24.7	23.9	22.8	22.8	23.5	24.5	24.8	25.4		
		MCDB	28.7	29.0	29.0	28.7	27.6	26.5	25.5	25.6	26.8	27.4	28.1	28.5		
	5%	WB	25.1	25.3	25.4	25.3	24.2	23.4	22.4	22.3	23.1	24.0	24.5	24.9		
		MCDB	28.2	28.6	28.6	28.2	27.0	26.0	25.1	25.1	26.0	26.6	27.5	28.0		
	10%	WB	24.7	25.0	25.0	24.9	23.7	22.9	22.0	21.8	22.6	23.6	24.1	24.5		
		MCDB	27.7	28.2	28.2	27.7	26.4	25.4	24.6	24.6	25.4	26.2	26.8	27.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.8	6.8	6.6	6.5	6.3	6.6	6.6	6.6	6.0	5.8	6.2		
		MCDBR	7.0	7.3	7.4	7.3	7.2	7.0	7.0	7.3	7.3	7.0	7.1	7.0		
	5% WB	MCWBR	2.8	2.9	2.9	3.1	3.4	3.4	3.6	3.6	3.3	3.0	2.8	2.7		
		MCWBR	6.6	6.7	6.9	6.8	6.6	6.4	6.6	6.7	6.7	6.3	6.3	6.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.392	0.387	0.386	0.382	0.372	0.361	0.356	0.372	0.423	0.426	0.419	0.411		
		taud	2.502	2.521	2.544	2.549	2.563	2.583	2.580	2.502	2.354	2.375	2.422	2.444		
	Ebn at Noon	Ebn at Noon	952	948	927	896	870	864	878	892	879	901	920	934		
		Edh at Noon	115	112	106	99	92	87	89	102	126	128	124	122		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.57	6.43	5.81	5.07	4.29	4.00	4.23	4.84	5.33	5.73	5.66	6.23		
	RadStd	RadStd	0.51	0.48	0.46	0.39	0.23	0.21	0.20	0.27	0.44	0.64	0.59	0.57		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page